

Special Practicals Jenkins- Part 2(Jenkins Controller-Agent Architecture on AWS EC2)

Jenkins / Nodes / ec2-agent / Log

```
UID=1000
USER=ubuntu
XDG_RUNTIME_DIR=/run/user/1000
XDG_SESSION_CLASS=user
XDG_SESSION_ID=7
XDG_SESSION_TYPE=tty
_='']'
[05/06/26 23:23:54] [SSH] Starting sftp client.
[05/06/26 23:23:54] [SSH] Copying latest remoting.jar...
Source agent hash is 2EC718000BE952A7E80F64AB21AEF921. Installed agent hash is 2EC718000BE952A7E80F64AB21AEF921
Verified agent jar. No update is necessary.
Expanded the channel window size to 4MB
[05/06/26 23:24:23] [SSH] Starting agent process: cd "/home/ubuntu/jenkins-agent" && java -jar remoting.jar -workDir /home/ubuntu/jenkins-agent -jar-cache /home/ubuntu/jenkins-agent/remoting/jarCache
May 06, 2026 5:54:24 PM org.jenkinsci.remoting.engine.WorkDirManager initializeWorkDir
INFO: Using /home/ubuntu/jenkins-agent/remoting as a remoting work directory
May 06, 2026 5:54:24 PM org.jenkinsci.remoting.engine.WorkDirManager setupLogging
INFO: Both error and output logs will be printed to /home/ubuntu/jenkins-agent/remoting
<==[JENKINS REMOTING CAPACITY]==>channel started
Remoting version: 3352.v17a_fb_4b_2773f
Launcher: SSHLauncher
Communication Protocol: Standard in/out
This is a Unix agent
Agent successfully connected and online
```

REST API

Jenkins 2.541.3

localhost:8080/computer/

Jenkins / Nodes

Nodes

+ New Node

Configure Monitors

Add, remove, control and monitor the various nodes that Jenkins runs jobs on.

S	Name	Architecture	Clock Difference	Free Disk Space	Free Swap Space	Free Temp Space	Response Time
	Built-In Node	Windows 11 (amd64)	In sync	38.32 GiB	8.86 GiB	38.32 GiB	0ms
	ec2-agent	Linux (amd64)	1.1 sec ahead	15.44 GiB	0 B	454.32 MiB	4447ms
	node1		N/A	N/A	N/A	N/A	N/A
Data obtained		32 min	32 min	32 min	32 min	32 min	32 min

Icon: S M L

Legend

←

→

↺

localhost:8080/computer/ec2-agent/configure

☆

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Jenkins

Nodes

ec2-agent

Configure

🔍

⚙️

👤

🗑️ Delete Agent

⚙️ Configure

📜 Build History

📈 Load Statistics

📄 Script Console

📄 Log

📄 System Information

🔌 Disconnect

Build Executor Status

0/1

ec2-agent

Description ?

Plain textPreview

Number of executors ?

1

Remote root directory ?

/home/ubuntu/jenkins-agent

Labels ?

ec2

Usage ?

Use this node as much as possible

Save

Apply

?

Jenkins

Nodes

ec2-agent

Configure

🔍

⚙️

👤

Launch method ?

Launch agents via SSH

Host ?

16.112.69.116

Credentials ?

ubuntu

+ Add

Host Key Verification Strategy ?

Non verifying Verification Strategy

Advanced

Edited

Availability ?

Keep this agent online as much as possible

Node Properties

Disable deferred wipeout on this node ?

Disk Space Monitoring Thresholds

Save

Apply

Jenkins

Nodes / ec2-agent / Configure

REST API Jenkins 2.541.3

Node Properties

☐ Disable deferred wipeout on this node

☐ Disk Space Monitoring Thresholds

☒ Environment variables

List of variables

Name

TMPDIR

Value

/home/ubuntu/jenkins-agent/tmp

+ Add

☐ Tool Locations

Save

Apply

Instance details | EC2 | ap-south-1

ap-south-2.console.aws.amazon.com/ec2/home?region=ap-south-2#InstanceDetails:instanceId=i-08785dfb46d43f543

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EC2Instancesi-08785dfb46d43f543

EC2

DashboardAWS Global ViewEvents

▼ Instances

InstancesInstance TypesLaunch TemplatesSpot RequestsSavings PlansReserved InstancesDedicated HostsCapacity Reservations

▼ Images

AMIsAMI Catalog

▼ Elastic Block Store

VolumesSnapshotsLifecycle Manager

▼ Network & Security

Updated about 1 hour ago

Instance ID

i-08785dfb46d43f543

IPv6 address

-

Hostname type

IP name: ip-172-31-4-232.ap-south-2.compute.internal

Answer private resource DNS name

IPv4 (A)

Auto-assigned IP address

16.112.69.116 [Public IP]

IAM role

-

IMDSv2

Required

Operator

-

Public IPv4 address

16.112.69.116 | open address

Private IP DNS name (IPv4 only)

ip-172-31-4-232.ap-south-2.compute.internal

Instance type

t3.micro

VPC ID

vpc-0b1c4f488d06f8d7f

Subnet ID

subnet-01d5cfff474718174

Instance ARN

arn:aws:ec2:ap-south-2:368999587946:instance/i-08785dfb46d43f543

Private IPv4 addresses

172.31.4.232

Public DNS

ec2-16-112-69-116.ap-south-2.compute.amazonaws.com | open address

Elastic IP addresses

-

AWS Compute Optimizer finding

Opt-in to AWS Compute Optimizer for recommendation | Learn more

Auto Scaling Group name

-

Managed

false

Details

Status and alarms

Monitoring

Security

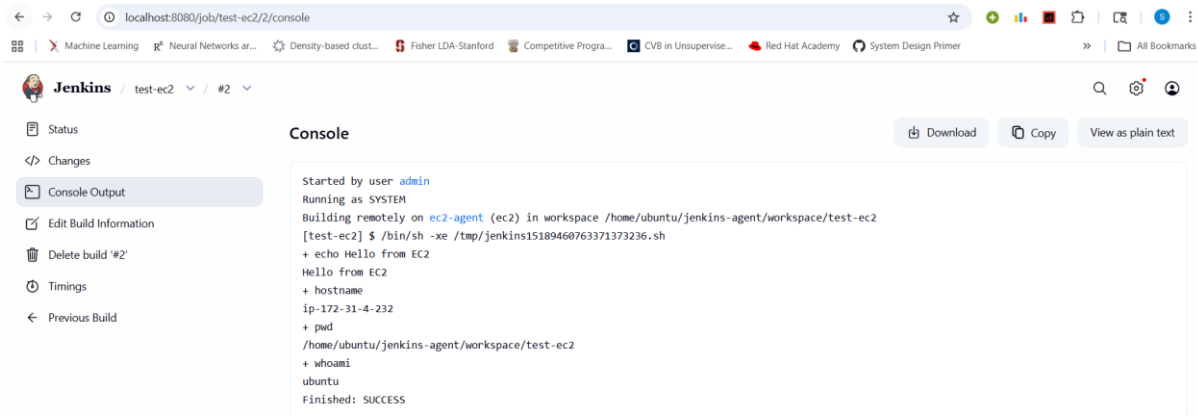
Networking

Storage

Tags

CloudShellFeedbackConsole Mobile App

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The screenshot shows a web browser at localhost:8080/jenkins. The Jenkins interface displays the console output for a build named 'test-ec2'. The output shows the build starting as 'admin', running as 'SYSTEM', and building remotely on 'ec2-agent (ec2)'. The console output includes the following text:

```
Started by user: admin
Running as SYSTEM
Building remotely on ec2-agent (ec2) in workspace /home/ubuntu/jenkins-agent/workspace/test-ec2
[test-ec2] $ /bin/sh -xe /tmp/jenkins15189460763371373236.sh
+ echo Hello from EC2
Hello from EC2
+ hostname
ip-172-31-4-232
+ pwd
/home/ubuntu/jenkins-agent/workspace/test-ec2
+ whoami
ubuntu
Finished: SUCCESS
```

Self-Notes:

Distributed Jenkins CI/CD Setup using AWS EC2 SSH Agents

- **Implemented a distributed Jenkins architecture with:**
 - Laptop as Jenkins Controller
 - AWS EC2 Ubuntu instance as Remote Agent
- **Configured AWS EC2 Security Group:**
 - Allowed only SSH (Port 22) access
 - Restricted SSH access using “My IP” for security
- **Connected to EC2 remotely using SSH:**
 - `ssh -i key.pem ubuntu@<EC2-Public-IP>`
- **Understood SSH key authentication:**
 - AWS stores public key in EC2
 - User authenticates using private .pem key
- Installed Java (OpenJDK 17) on EC2 because Jenkins agents internally run:

remoting.jar

- **Created dedicated Jenkins workspace on EC2:**

~/jenkins-agent

- Added EC2 as Jenkins Node using:
 - Launch Agents via SSH

- Username: ubuntu
- SSH private key credentials
- **Learned Jenkins Controller–Agent workflow:**
 - Controller schedules jobs
 - Agent executes builds remotely
 - Logs streamed back to Jenkins UI
- **Configured node labels:**
 - ec2

to force jobs to execute specifically on EC2 agent.

- **Faced /tmp low-space issue on EC2:**
 - Jenkins could not create temporary shell scripts
 - Node monitoring marked agent offline
- **Enabled:**
 - “Don’t mark agents temporarily offline” for:
 - Free Disk Space
 - Free Temp Space
- **Learnt difference between:**
 - Monitoring warnings vs actual execution failures
- **Created custom temp directory:**

/home/ubuntu/jenkins-agent/tmp

- **Used Linux environment variable:**

TMPDIR=/home/ubuntu/jenkins-agent/tmp

to redirect temporary file creation away from /tmp.

- **Understood environment variable inheritance:**
 - Jenkins Agent → Shell → Build Process
- **Applied Linux permissions:**

chmod 777 ~/jenkins-agent/tmp

to avoid permission-related temp file failures.

- Successfully executed remote Jenkins job on EC2:

echo "Hello from EC2"

hostname

pwd

whoami

- Verified distributed execution from Jenkins Console Output:

Building remotely on ec2-agent

Finished: SUCCESS